PREP 6 · LESSONS 11–12

# Power, mains and the National Grid

P = V I and P = I² R · energy transferred · mains electricity, safety and the Grid

SCORE / 10

Name _____ Class _____ Due _____

## Do Now REMEMBER

From earlier lessons · 1 mark each

1 Describe how the resistance of a thermistor varies with temperature. DN 105 [1]

2 Describe how the resistance of an LDR varies with light intensity. DN 106 [1]

3 Finding the I–V characteristic of a component: name the independent and dependent variables. DN 121 [1]

## Key ideas LESSONS 11–12

Show your working

4 A kettle is connected to the 230 V mains. The current is 8.0 A. Use P = V I

Calculate the power of the kettle. [2]

5 The current in a 15 Ω resistor is 2.0 A. Use P = I² R

Calculate the power transferred in the resistor. [2]

6 The frequency of the UK mains supply is | 50 Hz | 60 Hz | 230 Hz | [1]

7 State the function of the earth wire. [1]

8 Explain why step-up transformers are used before electricity is transmitted. [1]